

Federal Contract Oversight

FY2025 Risk Classification Framework

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Summary and key takeaways

The Federal Contract Oversight Challenge

\$700B+

The U.S. federal government awards over \$700 billion in contracts annually. With 600,000+ transactions, manual risk review is not feasible at scale.

Scale Impossible to Review

600,000+ transactions per year.
No officer can assess all contracts.
High-risk awards remain invisible.

No Standardized Risk Framework

Risk is multi-dimensional —
financial,
vendor, compliance, performance.
No unified system combines them.

Massive Financial Exposure

One undetected high-risk contract
can represent millions in
fraudulent
or wasteful government spend.

Background & Motivation

Why This Problem Matters

Federal procurement fraud and waste cost taxpayers billions annually (GAO, 2024).

Existing oversight relies on manual audit sampling — covers less than 5% of contracts.

No publicly available labeled dataset for federal contract risk exists.

Machine learning offers a scalable, data-driven approach to systematic risk detection.

Why Machine Learning

GAO procurement literature suggests 5–10% of federal contracts carry irregularity risk.

DoD, DHS, and HHS collectively represent the largest civilian and defense contracting spend.

USASpending.gov provides comprehensive FY2025 contract transaction data publicly.

Prior work in fraud detection (credit card, insurance) shows ML is highly effective on imbalanced tabular data.

Dataset — FY2025 Federal Contracts

600,862

Total Contracts

3

Departments
(DoD · DHS · HHS)

293

Raw Columns

FY2025

Data Period

Six Risk Dimensions Identified Across 56 Engineered Features

Financial Risk

Obligation amounts, value gaps, outlay ratios,
contract ceiling

Procurement Risk

Competition type, pricing type, award type,
number of offers

Compliance Risk

FedBizOpps posting, subcontracting plans,
cost pricing data

Vendor Risk

Foreign funding, SAM status, entity type,
business size

Performance Risk

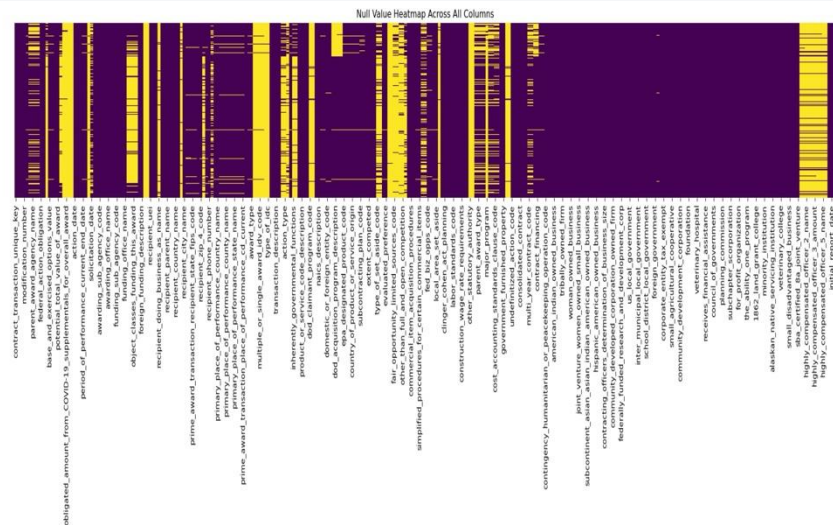
Contract duration, NAICS code, performance
dates

Socioeconomic Fraud

HUBZone, veteran, women-owned, 8(a) flag
combinations

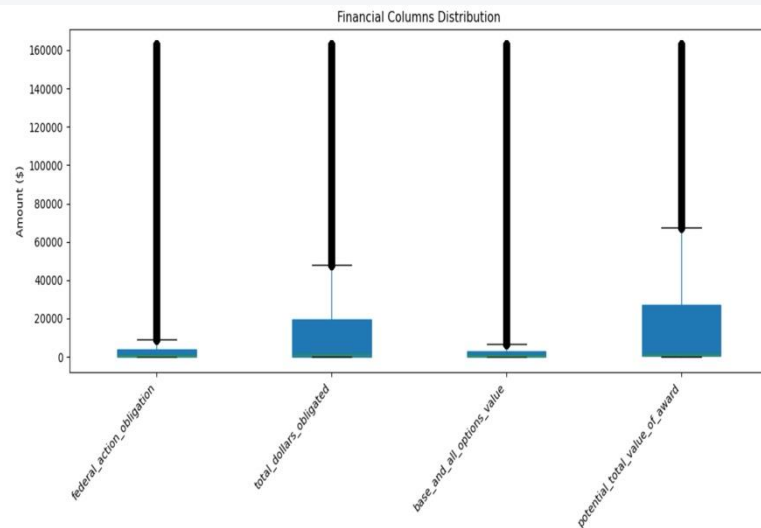
EDA — Data Quality & Financials

Null Value Heatmap — 293 Columns



Finding: Many columns are heavily null (yellow). Right-side columns (executive compensation, COVID supplements) dropped — not relevant to risk detection.

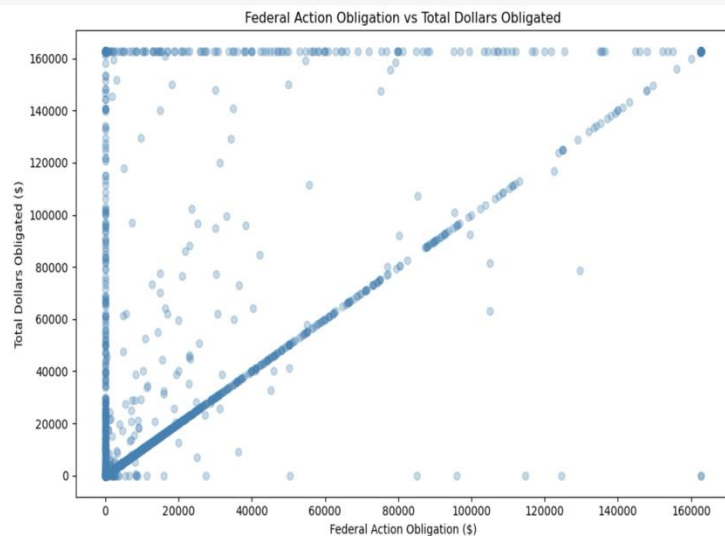
Financial Columns Distribution



Finding: All 4 financial columns are heavily right-skewed. Median near zero but outliers reach \$160K+. Log transformation applied before clustering.

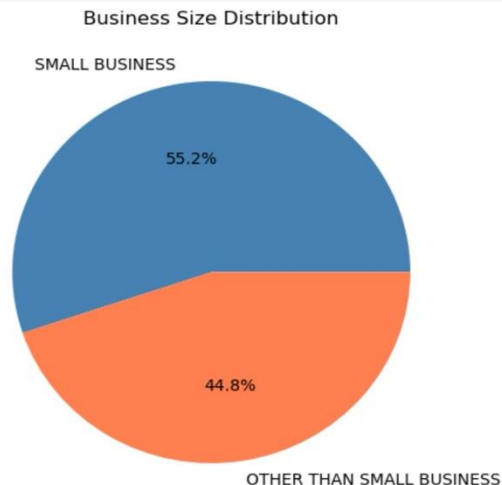
EDA — Correlation & Vendor Analysis

Obligation vs Total Dollars Obligated



Finding: Strong diagonal = most contracts have obligation $\approx$ total obligated. Off-diagonal outliers (top-left) = contracts where total obligated far exceeds the action amount — a financial risk signal.

Business Size Distribution



Finding: 55.2% Small Business vs 44.8% Other Than Small Business. Near-equal split raises misclassification risk — set-aside fraud is a key socioeconomic risk dimension.

Data Cleaning & Preparation

1

Load & Merge

Combined DoD + DHS + HHS CSV files → 600,862 rows, 293 columns
Dropped 4 fully null columns → 289 usable columns

2

Null Analysis

Identified columns with >50% null → dropped high-null columns
Selected 56 meaningful risk features across 6 dimensions

3

Date Engineering

Derived days_of_performance from start/end dates
Extracted modification_type prefix (STANDARD / PRICING / ADMINISTRATIVE)

4

Null Row Removal

Dropped rows with any remaining null across 56 features
464,385 clean rows retained (77.5% of original dataset)

5

Label Encoding

Label-encoded all 41 categorical columns
StandardScaler applied → X matrix (464,385 × 56)

Ground Truth — Three-Algorithm Majority Vote

No prior labels exist. Three structurally different unsupervised algorithms vote independently. A contract is labeled High Risk only when 2 of 3 agree.

K-Means

Groups contracts by feature similarity. Cluster with higher obligations and longer durations → High Risk.

62,293 flagged

k=2 · Silhouette 0.486

Isolation Forest

Scores each contract by how easily it can be isolated. Statistically anomalous contracts across any feature → flagged.

37,151 flagged

Contamination 8%

HDBSCAN

Hierarchical density-based clustering. Contracts with no dense neighborhood are labeled noise → High Risk.

21,172 flagged

min_cluster_size=100

Majority Vote Distribution

Rule: 2 of 3 flag as anomalous → HIGH RISK. All 3 agree normal → LOW RISK. Disagreement → UNCERTAIN, excluded from training.

K-Means	Isolation Forest	HDBSCAN	Label	Count
High Risk	Anomaly	Noise	HIGH RISK (1)	39,161
Normal	Normal	Normal	LOW RISK (0)	397,369
Disagree	—	—	UNCERTAIN — Excluded	27,855

39,161

HIGH RISK
Flagged

397,369

LOW RISK
Confirmed

27,855

UNCERTAIN
Excluded

436,530

TOTAL LABELED
For Training

Handling Imbalance, Validation & Hyperparameter Tuning

Class Imbalance — Class Weights

91% Low Risk vs 9% High Risk — naive model scores 91% accuracy by predicting everything Low Risk (useless). · Applied balanced class weights — High Risk errors penalized 10.15× more than Low Risk. · No SMOTE — class weights chosen for cleaner, noise-free training on real contracts.

10.15×

High Risk
penalty weight

Validation — 80/20 Stratified Split

Stratified split preserves 91/9 class ratio in both train and test sets. · Training set: 349,224 rows · Test set: 87,306 rows. · Stratification prevents evaluation bias from imbalanced splits.

80/20

Stratified
train/test

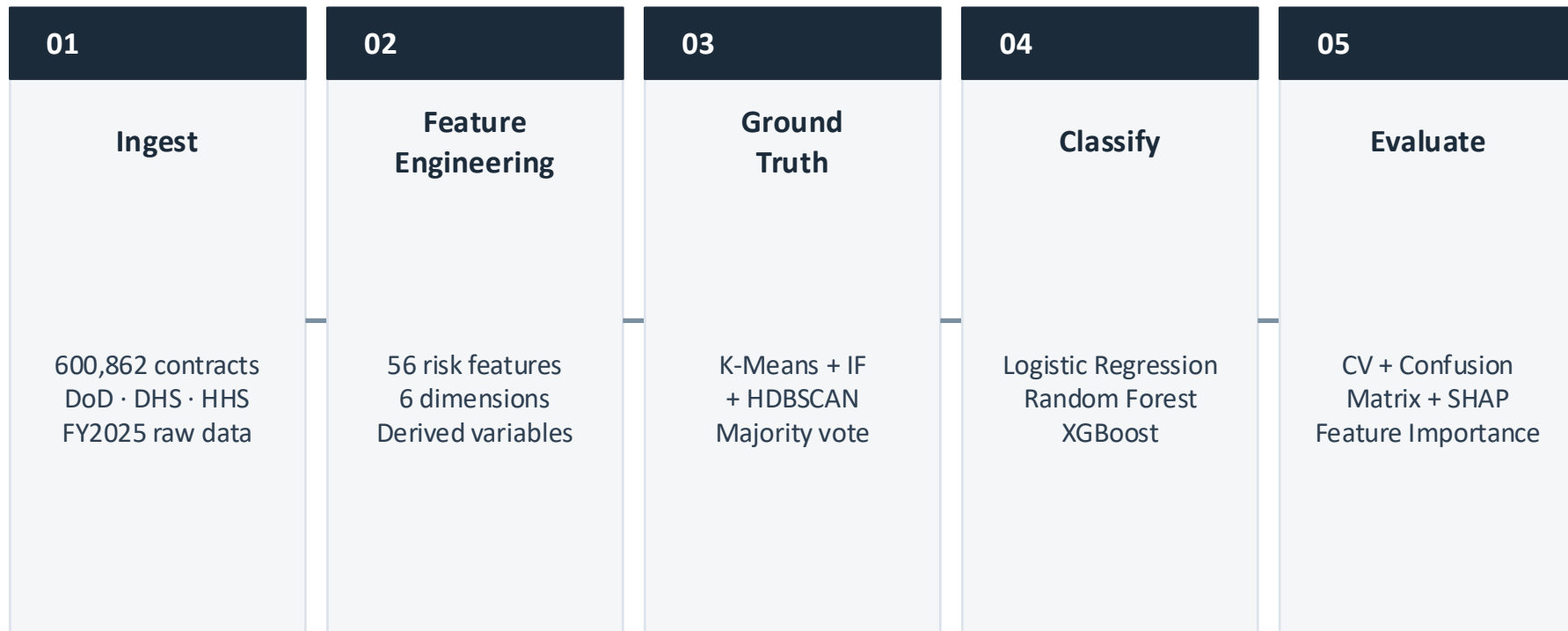
5-Fold Cross Validation

Repeated Stratified K-Fold (5 folds × 3 repeats = 15 evaluations). · Confirms model stability — XGBoost AUC 0.7574 ± 0.003 across all folds. · Low std confirms results are not dependent on any single data split.

0.003

AUC std
across folds

End-to-End Machine Learning Pipeline



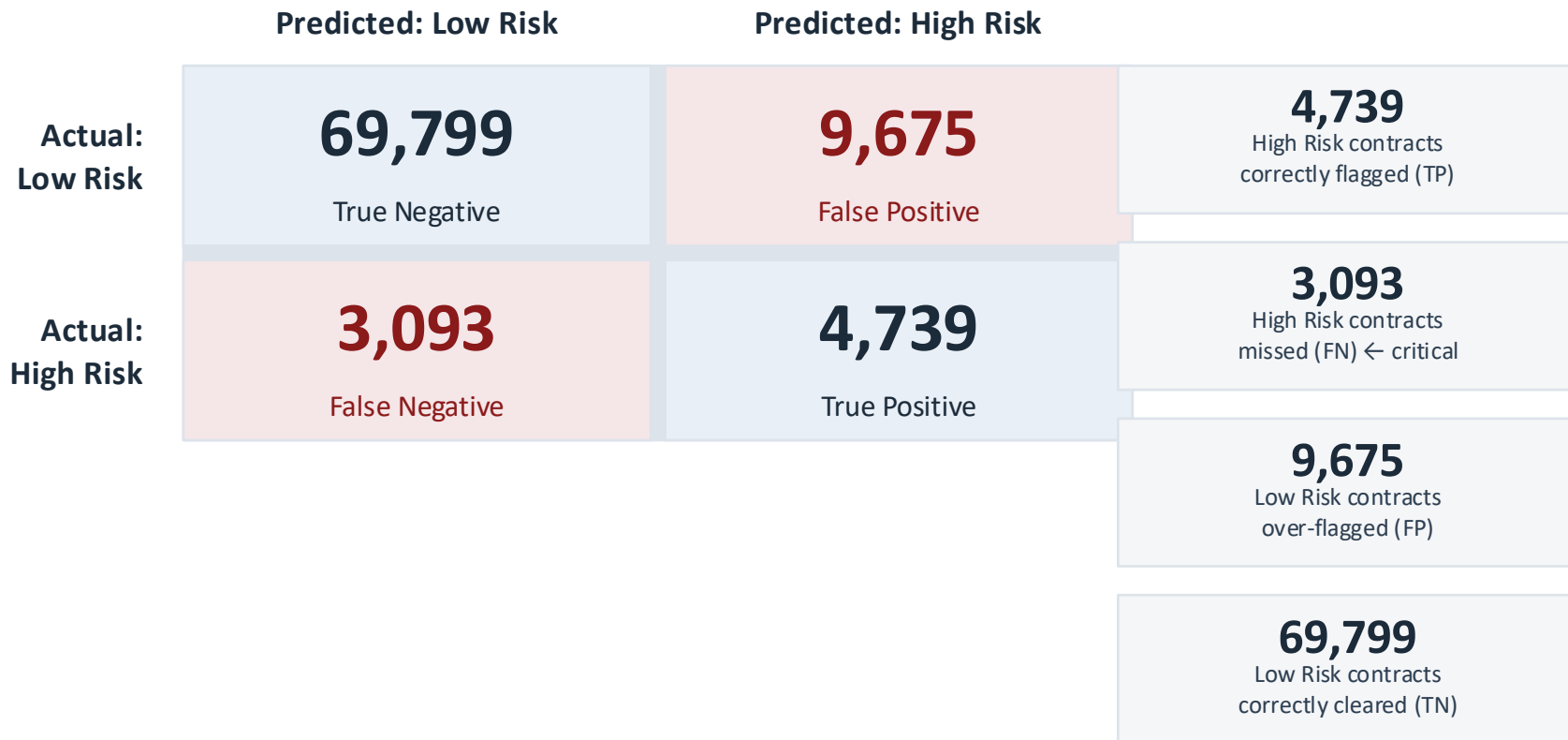
Classifier Performance Comparison

All models: 56 features · 80/20 stratified split · Class weights applied · Test set: 87,306 contracts

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression Interpretable baseline	76.0%	19.6%	54.1%	0.29	0.68
Random Forest Ensemble — 200 trees	90.6%	44.4%	17.6%	0.25	0.75
XGBoost Best performer ★	85.4%	32.8%	59.1%	0.42	0.7574

5-Fold CV — XGBoost: AUC 0.7574 ± 0.003 · Recall 59.1% ± 0.26% · F1 0.42 ± 0.002 → Stable across all folds

XGBoost — Confusion Matrix Breakdown



For federal oversight: FN (missed High Risk) is more costly than FP (extra review).

Top Risk Drivers — SHAP Analysis

Top 8 Risk Drivers — Mean |SHAP| (higher = stronger influence on High Risk prediction)

1	contract_financing		0.407	Non-standard financing (advance/progress payments) hardest to audit
2	product_or_service_code		0.231	IT and consulting categories carry higher inherent risk
3	days_of_performance		0.174	Longer contracts = more exposure, harder to monitor
4	total_dollars_obligated		0.159	Higher financial stakes correlate with High Risk
5	naics_code		0.153	Professional services and IT historically riskier
6	potential_total_value_of_award		0.118	Contract ceiling signals potential over-commitment
7	base_and_all_options_value		0.106	Options value gap between ceiling and obligated
8	base_and_exercised_options_value		0.098	How much was actually exercised vs awarded

Why XGBoost Outperforms — Class Weights Approach

XGBoost — Configuration

n_estimators	200
scale_pos_weight	10.15 (auto-computed)
Null handling	Native — no imputation
Class imbalance	Built-in weight parameter
Eval metric	AUC
Random state	42
CV folds	5-fold stratified

Why XGBoost Wins

1. Gradient Boosting

Iteratively corrects prior tree mistakes — more sensitive to minority High Risk class than Random Forest bagging.

2. scale_pos_weight

Penalizes High Risk errors 10.15× — directly addresses class imbalance in the loss function.

3. Tabular Design

Built for structured data — handles mixed feature types and missing values natively without preprocessing.

4. SHAP Explainability

Feature importance via SHAP shows which signals drove each contract's High Risk label — fully traceable.

Interpreting the Results

XGBoost ROC AUC 0.7574

Substantially better than random (0.50) and logistic regression (0.68). Confirms non-linear risk patterns exist in federal contract data.

Precision 32.8%

Of every 100 contracts flagged, 33 are genuinely High Risk. While not high precision, the priority for oversight is recall — missing a risky contract is more costly than reviewing a clean one.

Financial Features Dominate

7 of the top 15 SHAP features are financial — obligation amounts, contract values, and outlay ratios. This is consistent with GAO procurement audit findings.

Recall 59.1% on High Risk

Model catches 6 in every 10 genuinely High Risk contracts. For oversight purposes, this means 4,739 out of 7,832 High Risk contracts in the test set were correctly flagged.

Contract Financing as Top Driver

SHAP analysis reveals contract financing type is the dominant risk signal (0.407 mean SHAP). Non-standard financing arrangements are the strongest predictor of High Risk.

Stable Across CV Folds

AUC std of 0.003 across 5 folds confirms the model is not overfitting to a particular data split. Results are reliable and reproducible.

Success — What Was Achieved

Without This System

- 600,000+ contracts — no automated prioritization.
- High-risk awards buried — impossible to find manually.
- Fraud and waste signals go undetected for months.
- Resources wasted reviewing low-risk contracts.
- No consistent risk methodology across agencies.

With This System

- ✓ 39,161 High Risk contracts flagged automatically.
- ✓ 4,739 confirmed High Risk caught in test set (60.5%).
- ✓ Every contract traceable by Award ID and risk score.
- ✓ Oversight teams focus on the 8.4% that matters most.
- ✓ Unified risk framework applied across DoD, DHS, HHS.

Novel Contributions to the Field

1

Three-Algorithm Majority Vote Ground Truth

First application of K-Means + Isolation Forest + HDBSCAN majority vote for constructing defensible ground truth labels in the absence of human-annotated federal contract risk data. Addresses a fundamental gap in public sector ML.

2

Comprehensive FY2025 Federal Contract Risk Framework

End-to-end risk detection pipeline across DoD, DHS, and HHS covering 600,862 contracts. Six risk dimensions — financial, procurement, compliance, vendor, performance, socioeconomic — operationalized as ML features.

3

SHAP-Based Explainability for Procurement Oversight

Application of SHAP (SHapley Additive exPlanations) to federal contract risk detection provides contract-level explainability. Oversight officers can see exactly which features drove each High Risk prediction.

4

Contract Financing as Dominant Risk Signal

Novel empirical finding: contract_financing type is the single strongest predictor of High Risk (SHAP0.407), surpassing financial obligation amounts. This has not previously been quantified in federal procurement ML literature.

Future Work & Limitations

Human-Annotated Validation Set

Collaborate with GAO or IG offices to obtain independently labeled contract risk data for genuine out-of-sample validation beyond the self-supervised ground truth.

Temporal Generalization

Train on FY2024 data and test on FY2025 to evaluate whether risk patterns generalize across fiscal years and procurement cycles.

Cross-Agency Deployment

Extend pipeline to all 15 federal departments. Current model trained on DoD, DHS, HHS — test transferability to other agencies (Treasury, DOE, DOT).

Real-Time Risk Scoring

Integrate with USASpending.gov API for live contract risk scoring as new awards are posted. Enable proactive rather than retrospective oversight.

Hyperparameter Optimization

Apply Bayesian optimization (Optuna) to tune XGBoost hyperparameters systematically. Current `scale_pos_weight` is computed from class ratio, not optimized.

Ensemble of All Three Classifiers

Combine LR + RF + XGBoost predictions via soft voting or stacking. LR's high recall (54%) and XGBoost's high AUC (0.76) may complement each other.

Conclusion

We built an end-to-end ML pipeline that automatically identifies High Risk federal contracts from 600,862 FY2025 transactions across DoD, DHS, and HHS.

A novel three-algorithm majority vote (K-Means + Isolation Forest + HDBSCAN) was used to construct defensible ground truth labels without human annotation.

XGBoost with class weights achieved ROC AUC 0.7574 (± 0.003) and recall of 59.1% — the best performance across Logistic Regression, Random Forest, and XGBoost.

SHAP analysis identified contract financing type as the dominant risk driver, followed by product/service code, days of performance, and total dollars obligated.

The system flagged 39,161 High Risk contracts (8.4% of 464K labeled contracts), providing a ranked, traceable list for oversight teams to act on.

This work demonstrates that unsupervised ground truth construction combined with supervised classification is a viable and scalable approach to federal contract risk detection.

Reference List

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Thank You

Questions & Feedback